The rank effect of trade on Brazil's foreign policy political alignment with China

Manoel Galino

2024-06-30

Abstract

This paper investigates the causal effect of China's rise to Brazil's top trade partner status on Brazil's foreign policy alignment with China. Using a synthetic difference-in-differences (SDiD) approach, the study leverages a "rank effect" — the change in salience when China became Brazil's largest trade partner in 2009 — as a quasi-natural experiment. We measure foreign policy alignment through ideal point estimates from UNGA voting data (Bailey et al., 2017) over the period 1991–2015. The findings suggest that surpassing historical trade partners triggers a qualitative shift in political alignment, decreasing the ideological distance between Brazil and China at the UNGA by 39%.

Word count: 5139

1 Introduction

In 2009, China surpassed the US as Brazil's biggest trade partner. The economic ties between both countries only grew over time, including the relevance for China exports. In 2004, Brazil was the 15th largest market for China and became the eighth in 2014. The foreign policy of both countries also showed some alignment. Both countries became members of forums such as G5 (Brazil, Mexico, India, China and South Africa), BASIC (Brazil, South Africa, India and China) and the BRIC summit. The theme of China's influence due to its economic integration is a major theme in International Relations. A report from the US Defense Department argued in 2018 that : "China intends to use [Belt and Road Initiative] to develop strong economic ties with other countries, shape their interests to align with China's…" (U.S. Department of Defense 2018, x). In fact, a major strand of the literature argues that expanding trade ties with China had significant foreign policy consequences for Brazil and other Latin American countries (Flores-Macías and Kreps 2013; Kastner and Pearson 2021).

Nonetheless, things are more complicated than it seems at first sight. On one hand, it is easy to point out how economic ties with China changed Brazil foreign policy, a major example being the recognition of China as a market economy by Brazil in 2004, before most Latin American Countries. In a study about the prevalence of trade in Brazil's president Lula's speech on China, (Guilhon-Albuquerque 2014) shows that that theme was emphasized during Lula's second term. On the other hand, with deeper economic ties than during Lula's first term, Bolsonaro's term saw a distancing from China and a closer alignment with the US. The economic integration was cast as dependence and created a backlash effect. And under Lula's presidency after Bolsonaro, Brazil has not yet accepted to participate in the Belt and Road Initiative, despite speculations to the contrary.

In fact, other studies have argued for the reverse causal order. They contend that complementary aligned national interests and shared identities with the "emerging" discourse (Haibin 2010) and a revisionist stance towards the international order promoted more trade links between countries (Coutinho and Rodriguez 2024). In fact, many studies have found evidence that political alignment with China in other contexts and periods can significantly impact trade flows with it (Fuchs and

Klann 2013; Aiyar, Malacrino, and Presbitero 2024; Yan and Zhou 2021).

Opposite causal pathways make it hard to deal with the problem of reverse causality and omitted variable bias in observational studies. As a result, we still need to understand better the effect of trade on political alignment. Moreover, many of the existing studies do not differentiate exactly what is the economic mechanism: is it its foreign direct investment in the region? The importance of China's imports to the economic success of the Latin American countries? Are there other non-economic factors, such as China's growing soft-power outreach or shared claims for reforming global governance institutions and resistance to Washington-centric norms that are important? Recently some studies have argued that it is a mixture of many of these forces working at the same time (Struver 2014; Schenoni and Leiva 2021).

The present study aims to deal with those two order of problems (causal credibility of findings and disentangle the effect of trade from other effects). We do so leveraging a discontinuity in the rank of an otherwise continuous variable: amount of trade with a given country. We consider that becoming the first trade partner in 2009 is the treatment that caused salience in Brazil. We use a synthetic difference in differences approach to estimate the causal effect of China becoming the first, as opposed to second, trade partner for Brazil to estimate its causal effect on foreign policy orientation, measured as the absolute distance in UNGA ideal point estimation (Bailey, Strezhnev, and Voeten 2017) between Brazil and China over 1990-20014. We found that the change in the rank position decreased the distance by -.29 points. To put this number in context, the average country in our sample had a mean absolute distance of .84 point, and Brazil had an average absolute distance of 0.75 points before 2009, which means that we are estimating a decrease of 39% in the ideal point distance between Brazil and China after 2009.

Our results show that when salient, the trade position of a country has a distinctive causal effect on the foreign policy of a country like Brazil. All the arguments and discussion about China as an emerging power overlooked the importance of becoming salient in a qualitative distinct way to impact the foreign policy of other countries. It is not only a matter of having more trade ties, but overcoming historical trade partners that matter. This result also points to the need to investigate similar salience, rank effects, in other economic variables (such as Foreign Direct Investment and Loans) and variables along other dimensions (such as diplomatic ties, frequency of bilateral meetings and deals and agreements signed).

The rest of the paper is organized as follows. Next, we discuss the importance of salience effects in the realm of trade effects and political alignment. Then, we evaluate how to measure political alignment, our main outcome variable. In the fourth section of the paper, we present our methodology, data and variables. Then, we present our empirical findings. The last section presents our concluding remarks.

2 Trade and Political Alignment

Some studies argue that the trade effects are due to coercive capacity to threaten countries to behave in the ways China wants (Tanner 2007; Reilly 2013). Other studies suggest that deepening economic ties creates vested interest in which economic groups would lobby on behalf of China due to their self-interest (Kastner and Pearson 2021). Another argument is that economic integration can transform public and elite opinion over China, either via soft power (Morgan 2019) or through the way media reports on China, changing public perceptions (Kastner and Pearson 2021). Finally, there is the dimension of structural power. China's diplomatic inroads into Latin America opened up the possibility of more autonomy from the US in the region (Struver 2014). Schenoni and Leiva (2021) argue that recent Brazilian foreign policy change is the result of the pull of China as a new great power, resulting in what they call a "dual hegemony". According to the ground break work by Neto (2011), as Brazil became a more powerful country, it could distance itself from the US. It is thus expected that as China emerged as an important country, it would attract some alignment of Brazilian foreign policy due to this effect of power. Mourón and Urdinez (2014) argued that the power gap was the most important factor, and that it was true for the broader South American countries: the smaller the power gap between a country and the U.S., the less it aligned with the U.S. at the UN.

Qualitative studies have also pointed to the linkage between trade and foreign policy. In a study about the prevalence of trade in Brazil's president Lula speech on China, (Guilhon-Albuquerque 2014) shows that trade is the thematic emphasis of Brazil foreign policy during Lula's second term. From the other side, Vadell (2011) emphasizes how the diplomatic ties between China and South America countries followed the economic interests of the Asiatic power in the region.

Other studies have emphasized the reverse causal order. They contend that complementarity of aligned national interests and shared identities with the emerging discourse (Haibin 2010) and a revisionist stance towards the international order promoted more trade links between countries (Coutinho and Rodriguez 2024). Many studies have found evidence that political alignment (or its absence) can significantly impact trade flows. Countries that bucked China's political preferences – by officially receiving the Dalai Lama – suffered subsequent trade declines (Fuchs and Klann 2013). Countries that voted in support of the PRC (political alignment) in 1971 on the matters of Taiwan enjoyed a surge in bilateral trade with China in the immediately following year (Yan and Zhou 2021). In a more general quantitative study, Chen and Zhou (2021) shows that a country tends to import more from partners with whom it has friendly relations, which is especially true for an authoritarian state (such as China). Studies have found a similar effect in other economic variables, such as Foreign Direct Investment (Aiyar, Malacrino, and Presbitero 2024).

The most similar study to ours in terms of causal ambition is the pioneering quantitative study of Flores-Macías and Kreps (2013). They show that the more countries trade with China, the more likely they are to side with China on foreign policy, in which the main measure of political alignment is vote similarity at UNGA. In order to control for endogeneity, the authors used two distinct approaches: a fixed effect model with lagged variables and two-stage least squares. The first model is just wrong from a causal point of view, since one key identification assumption in fixed effects models is no carry over effects, which precludes the use of lagged dependent variable in the regression (Blackwell and Glynn 2018; Imai and Kim 2019, 2021). The instrument used in the latter approach is lagged energy production of a country. This variable, however, fails to meet the exclusion restriction assumption needed for a valid instrument, since energy production is, for example, related to economic development, which may be a causal pathway from the instrument to the outcome (political alignment). In general it is very hard to justify the exclusion restriction of instruments (Mellon 2021; Bastardoz et al. 2023; Lal et al. 2024), and this is no exception, diminishing the credibility of the assumptions needed for causal identification.

Another limitation of the literature is that all quantitative studies that measured Brazil's foreign policy alignment with UNGA votes used some similarity measure based on percentage of votes at UNGA (Neto 2011; Flores-Macías and Kreps 2013; Mourón and Urdinez 2014). However, as

Bailey, Strezhnev, and Voeten (2017) shows, such votes cannot be used in a panel data set because such indicators do not separate changes in the UN's agenda from shifts in state preferences. As a result, many attributions of changes in the policy orientation can in fact be changing in the content of the votes. Bailey, Strezhnev, and Voeten (2017) developed an alternative measure, based on ideal point estimation, that disentangle the effect of agenda changing from the foreign policy preference change. By using the estimates developed by Bailey, Strezhnev, and Voeten (2017), we will be able to consider only state preference reorientation over time, which is key for the purpose of the paper.

While important, the idea that economic ties with China can alters the perception and beliefs of political elites often overlooks how decision-makers process information about economic relationships Recent work in cognitive and behavioral sciences suggests that the salience of economic ties — not just their material depth — influences attention and policy choices. In the next section, we introduce the concept of the 'rank effect' as a mechanism capturing this overlooked cognitive dimension in the trade-foreign policy nexus.

3 Political effects of rank

Decision makers live in a complex, noise world. In such situations, it is ecologically rational to use cues from the environment to choose what to pay attention to and react accordingly [citation needed. Gingerezer]. Many studies in international relations on policymakers have emphasized the roles of cues and heuristics when making a foreign policy decision or reorientation.

A recent literature in the intersection of cognitive and social sciences has emphasizing the role heuristics in decision-making, like how redundant information nonetheless work by shifting one's attention (Conlon 2023), that people neglect correlated structure of information and treat them as independent (Enke and Zimmermann 2017) or the existence of behavioral attenuation, in the sense that, due to information-processing constraints, people fail to corrected update beliefs and make decisions in response to change in the fundamentals of a problem (Enke et al. 2024).

In the field of International Relations, the poliheuristic theory is a standard account on how decision makers also behave like regular people in this regard (Mintz 2004, 2005). On of the main arguments is that, due to information-processing constraints, poliheuristic decision-making process is based on a non exhaustive search in the sense that "comparisons of policy options are made within a (very) restrictive alternative set and attribute set" (Mintz and Geva 1997, 84). Consequently, policy dimensions or issues that attract attention may newly enter decision-makers' consideration sets, even if they were previously ignored.

The rank effect is not new in the political science literature and has been documented in domestic politics before (Anagol and Fujiwara 2016; Folke, Persson, and Rickne 2016; Granzier, Pons, and Tricaud 2023). It helps to coordinate decisions and produce some bandwagon effect on voters who want to vote for the winner. However, as far as we know, no one discussed rank effects in international politics, although they are just as likely to be real, as argued above.

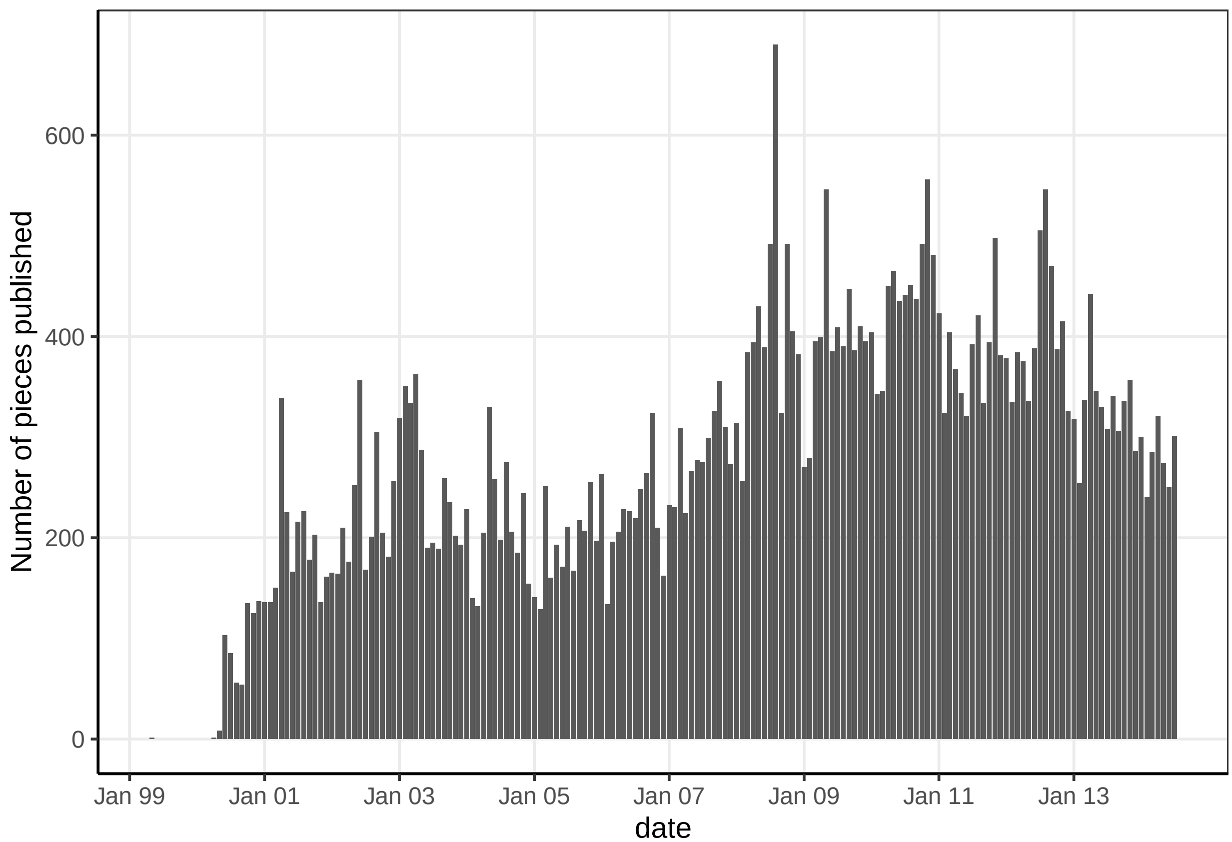

Figure 1: Number of pieces published with term 'China' at newspaper Folha de São Paulo.

It is quite clear from figure xx that upt to 2006, the main themes covered about China was related to health issues, North Korea and nuclear issues, and some sports themes ("seleção brasileira"). In 2009, those themes continued to be covered by the press, but much more news about Asia, China and Hong Kong stock market, along with a central network of news piece about brazilian company

Vale do Rio Doce, a major exporter of raw products to China, connected to the Dollar, always an issue for the Brazilian trade. This evidence is central to the mechanism we are proposing here. The economic ties are qualitatively different between brazilian companies, its economy and China that it show up in the press. It is useful to zoom in and compare 2008 with 2009.

We must remember that 2008 was the year of the financial economic crisis known in the US as the Great Recession. It is expected that, givenm the growing importance of China for the global economy at the time and for Brazil, that news about the stock market and the chinese economic will be present. However, it is critical for our argument that there is no news about Brazilan or Chinese companies, not any mention about trade in the news in a recurrent fashion. Thus, we filtered trigrams with frequency higher than five whem comparing 2008 with 2009.

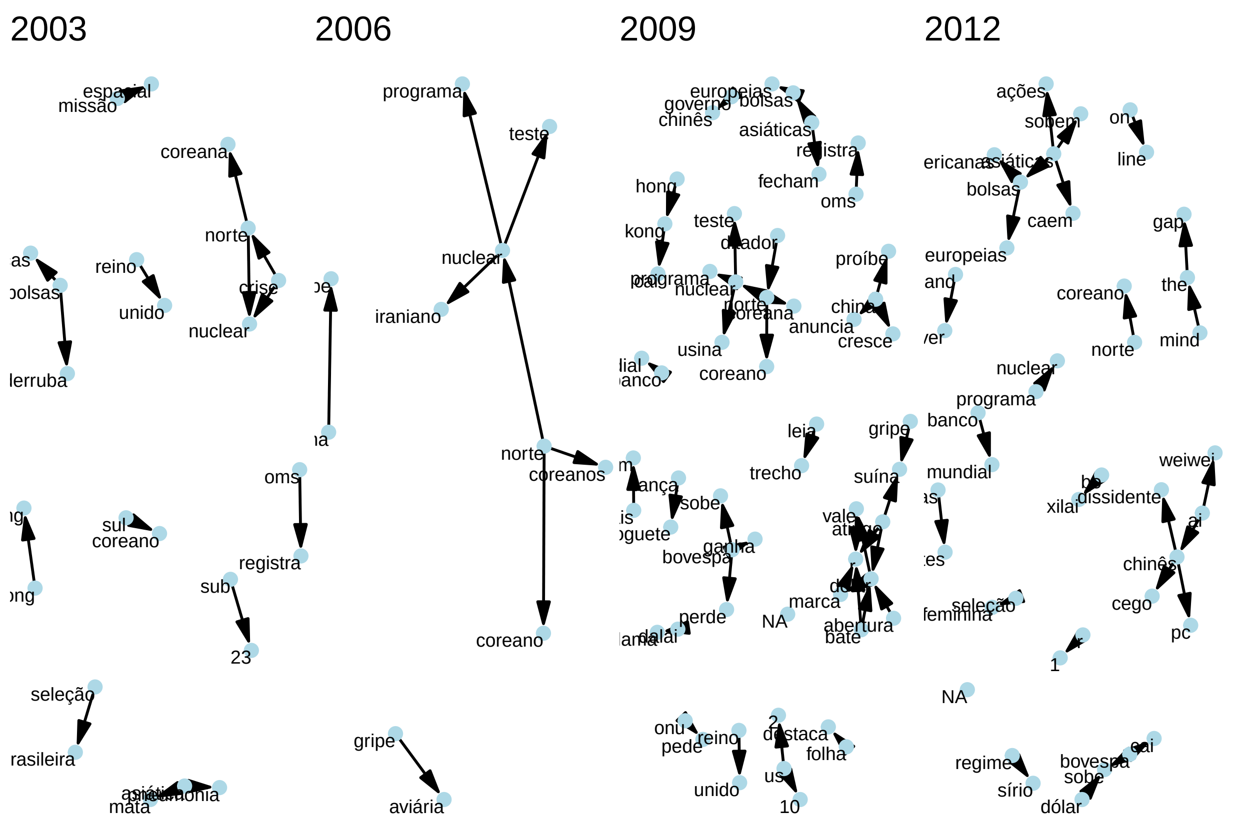

Figure 2: Trigram of pieces published about China at newspaper Folha de São Paulo. Darker arrows are more frequent trigrams. Only trigrams with co-ocurency frequency higher than 8 are shown. The low number of trigrams in 2003 and 2006 is due to the lower number o pieces published.

In order to establish that a rank effect happened around the time we are discussing, we collected all stories, ope-ed and editorials with the term “China” published at the newspaper Folha de São Paulo, the paper with widest circulation in Brazil [citation needed]. The results include some false positives (as a piece involving a football player nicknamed “China”), but they are rare. In a random sample of 100 stories we found no instance of a false positive. Also, the stories cover many different themes, such as sports, natural disasters, politics, economics, science and so on. Nonetheless, they are a measure of interest about everything related to China. Over time, news pieces about China grew up to a peak in 2008, the year of the Summer Olympics in China and the year when China was overcoming the US as Brazil’s first trade partner. A good indicator to measure a change in tone is to compare the amount of news in January of each year, since it is a time when there is congress recess, just before carnival and not much news (apart from the first year of a president, like 2003, 2007 and 2011). Thus, by looking at this month every year, we can see that 2008 is indeed when there is a break from before.

Here are some headlines with “bate” (meaning “reach”)^[I have used chatgpt 4o to assist in the headlines translation from Portuguese into English. See https://chatgpt.com/share/6862b919-4630-800e-ab9c-df9dee5ae56c]: In January 9th, 2009, “Exportações do agronegócio batem recorde, com destaque para China” (Agribusiness exports reach record high, with China as the main highlight). On April, “China bate recorde de importações de minério de ferro em abril” (China hits record iron ore imports in April). On August: “Vendas de minério do Brasil para China batem recorde em julho” (Ore sales from Brazil to China reach record high in July) and so on. A key formulation using the word “bate” (hits or reach record high) in an economic context appeared many times in 2009, and only one in 2008, about the agribusiness.

What about the content over time? If our hypothesis is corrected, not only we should expect more pieces abolut China, but also a growing number of pieces about economics and trade in particular. In order to assess, we computed trigrams of each headline for four years (2003, 2006, 2009 and 2012) to see how they evolve over time. By looking at the most frequent trigrams, we can have a sense of the what kind of content was in each year.

If, as the we argued, symbolic milestones like surpassing historical trade partners attract disproportionate attention from policymakers, then the moment China became Brazil’s largest trade partner constitutes a plausible, exogenous ‘shock’ to the salience of the bilateral relationship. This logic

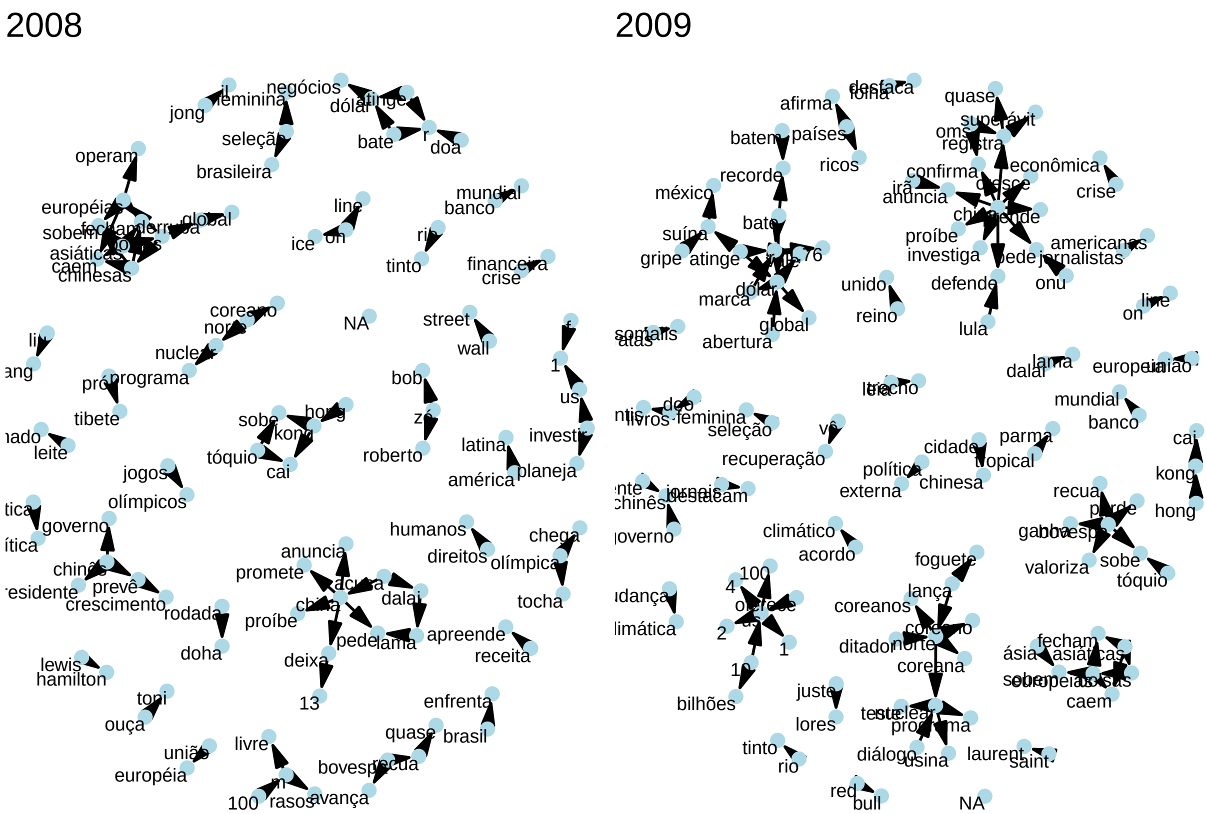

Figure 3: Trigram of pieces published about China at newspaper Folha de São Paulo in 2008 and 2009. Only trigrams with co-ocurency frequency higher than 5 are shown.

motivates our empirical strategy, which leverages this rank change as a quasi-experimental treatment using a Synthetic Difference-in-Differences approach, as detailed in the next section

4 Methodology

We explore a discontinuity in the change of China as the second versus first Brazil trade partner to estimate the (local) causal effect of increased importance of China becoming the second largest trade partner of Brazil. This allows us to approximate a difference-in-differences methodology (DiD). As we know, the key identification assumption in DiD is the so-called parallel trends assumption, which states that, absent the intervention, the treated unit (Brazil) would behave just as the average of the control group of countries. This assumption is not credible in the present case, because many things happened during this period (like the 2008 economic crisis and its economic and political effects in the following years), which breaks the credibility that countries would respond the same way to this shock, leading to bias in the estimate of the causal effect.

In order to avoid these problems, we employ the synthetic difference in differences (SDiD) method, developed by Arkhangelsky et al. (2021). It combines the DiD estimator with the synthetic control method (SCM). It provides a more flexible approach than both methods separately. On one hand, DiD methods do not work well with a single treated unit (inflating he standard error) and they also rely on the parallel trends assumption for identification of the causal effect. On the other hand, SCM tries to match the pre-treatment trend in the outcome based on covariates to estimate the counterfactual outcome. However, it compares the whole time period before and after the intervention and does not allow the inclusion of unit fixed effect, with can be a cause of bias, specially in the case of slow moving variables. SDID try to solves both problems (non-parallel trends for DiD, time period and absence of fixed effects for SCM) by combining both methods. Like in the SCM model, we are still trying to estimate the causal effect comparing the post-treatment behavior of the synthetic control and the treat unit, buu with steroids, so to speak. The specific model is as follows:

Consider a complete panel[1]: N countries measured over T years, the outcome variable for country i in time t is Y_{it} and the binary treatment variable $D_{it} \in \{0, 1\}$. Suppose also that the first $N_{\text{countries}}$

[1] sometimes called a balanced panel. We prefer the term "complete" to avoid confusion with balancing of covariates, which is a standard analysis in causal studies

do not receive the treatment and the last country N_{Brazil} do receive the treatment at t_0. The job is, like in SCM, to find weights $\widehat{w}^{\text{SDiD}}$ such that before t_0 treated and control countries match as closely as possible, that is, $\sum_{i=1}^{\text{countries}} \widehat{w}^{\text{SDiD}} Y_{it} \approx \sum_{i=1}^{\text{Brazil}} Y_{it}$, for all $t = 1, ..., t_0$. Differently from SCM, we do not stop here. We then find $\widehat{\lambda}_t^{\text{SDiD}}$ to balance the time trend before and after t_0. Lastly, the average causal effect on the treated τ_{ATT} is estimated by a two-way fixed effect regression model (as in standard DiD models):

$$(\widehat{\tau}_{ATT}^{\text{SCM}}, \widehat{\mu}, \widehat{\beta}) = \underset{\tau_{ATT},\mu,\beta,\alpha}{\arg\min} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT}\right)^2 \widehat{w}^{\text{SDiD}} \widehat{\lambda}_t^{\text{SDiD}}$$

Vale a pena contrastar esse modelo com o DiD clássico e o SCM. O modelo de DiD estima:

$$(\widehat{\tau}_{ATT}^{\text{SCM}}, \widehat{\mu}, \widehat{\lambda}, \widehat{\beta}) = \underset{\tau_{ATT},\mu,\alpha,\beta}{\arg\min} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT}\right)^2$$

Como se vê, o SDiD coloca pesos em países dos grupos controle que são mais similares ao país do tratamento e mais peso em períodos pré-tratamento que são mais similares ao período pós-tratamento. Por outro lado, o SC estima um modelo como:

$$(\widehat{\tau}_{ATT}^{\text{SCM}}, \widehat{\mu}, \widehat{\beta}) = \underset{\tau_{ATT},\mu,\beta}{\arg\min} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \beta_t - D_{it}\tau_{ATT}\right)^2 \widehat{w}^{\text{SDiD}} \widehat{\lambda}_t^{\text{SDiD}}$$

As we can see, there is no unit fixed effect and no time weight. As a result, the SCM cannot use unit unobserved invariant characteristics to approximate the counterfactual outcome and will consider the whole time period when estimating the causal effect. This is a problem for our data, because the relationship between China and Brazil becomes more entangled as times goes by and years much further from the treatment period are likely not a good comparison to periods long before the treatment.

Another key advantage of the SDID is that it reduces the variance of the estimator, allowing more precise estimation (Arkhangelsky et al. 2021). One problem with this model, however, is that it does not include time-varying covariates that are important to more closely match the pre-treatment time period. Thus, to account for time-varying covariates and better adjust the model, we include vector of time-varying controls as suggested by (Hirshberg and Klosin 2024), such that the estimated equation is now:

$$(\hat{\tau}_{ATT}^{\text{SCM}}, \hat{\mu}, \hat{\beta}, \hat{\gamma}) = \underset{\tau_{ATT},\mu,\beta,\alpha,\gamma}{\arg\min} \sum_{i=1}^{N}\sum_{t=1}^{T}\left(Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT} - X_{it}\gamma\right)^2 \hat{w}^{\text{SDiD}}\hat{\lambda}_t^{\text{SDiD}}$$

4.1 Data and Variables

Since we need to fit a model to a complete panel, we constraint the sample to 105 countries over a period of 21 years, from 1991 to 2011.

Our outcome variable, vote similarity between a country and China at UNGA, is based on the ideal point estimates based on UNGA vote data from Bailey, Strezhnev, and Voeten (2017). Other variables include per capita GDP from Penn World Table, version 10.01 (Feenstra, Inklaar, and Timmer 2015). Trade data are drawn from the International Trade and Production Database (Borchert et al. 2021). To avoid double counting, I use the value reported by the importer country to compute total trade between two dyads, since importers have more interest in measuring this value accurately for tax collection reasons. We use the share of total trade with China and the share of total trade with the United States as our two measures of trade. As our measure of power, we used the Global Power Index (GPI) from Diplometric – Pardee Institute, University of Denver. GPI combines demographic, economic, military, technology and diplomatic dimensions and then we computed the power gap between Brazil and the US. We also include distance to the US, as countries closer to the US will have a bigger incentive to align its foreign policy with it. Another economic variable is the exchange rate, since a complaint of China is its devalued currency. A country with a more devalued currency is expected to be more aligned with China than not. The definitions and measurements of key variables are provided in table 1. Table 2 presents the descriptive statistics.

5 Descriptive Analysis

In the figure below, we plot the evolution of UNGA's ideal point of US, Brazil and China over time. As we can see, Brazil is closer to China than to the US since the beginning of the series of my data, in 1990 up to 2022, during Bolsonaro's government. Even though there is a distance of Brazil under Bolsonaro's term, it is still closer to China than to the US. This is a testimonement of how much Brazil is structurally closer to China at the UNGA.

It also shows that Brazil is moving more over time than China, which is mostly stable over time. Thus, that is an indication that Brazil is moving closer to China than the opposite, as expected, given the differential of power and importance of both Countries and the dependence between them.

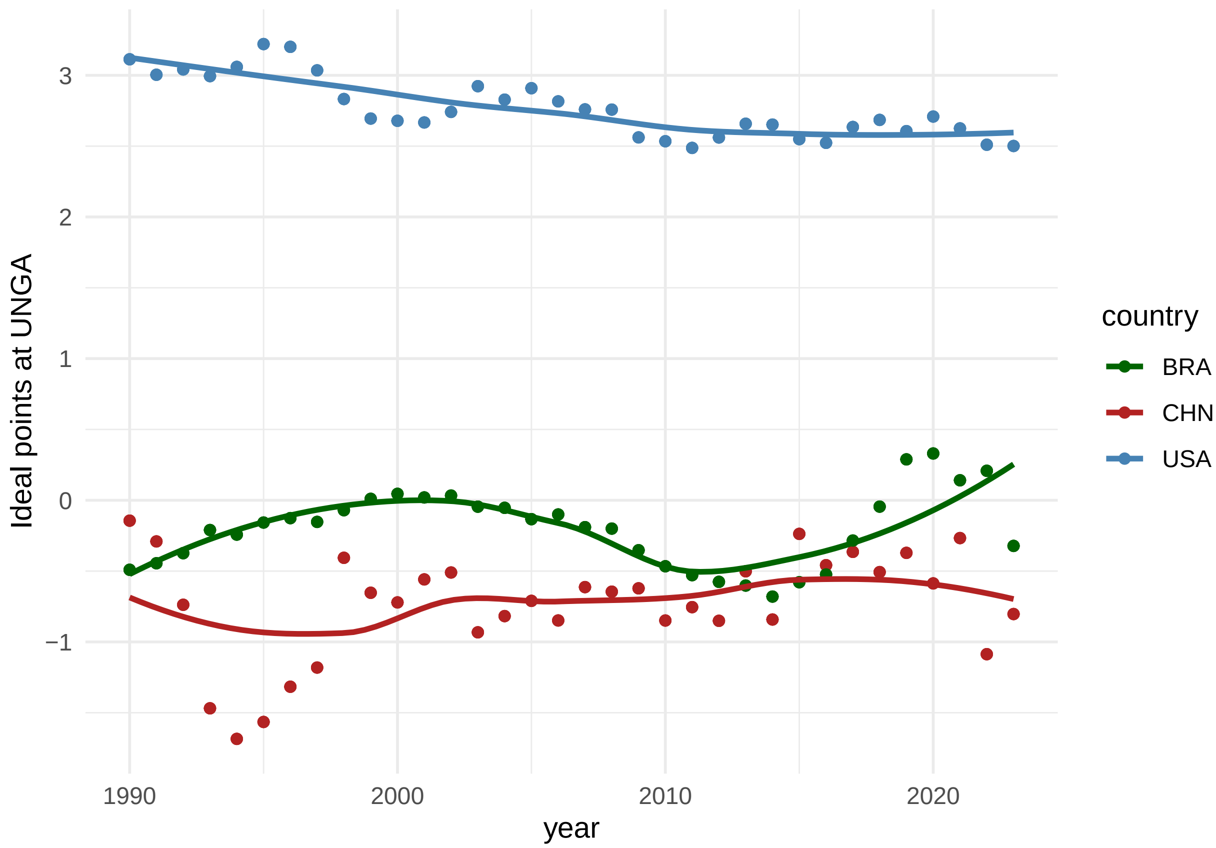

Figure 4: caption

Having established that it is mostly Brazil moving around than China, we can use as our main variable the absolute distance between Brazil and China's ideal point, with the understand that, a part from the beginning of the nineties, Brazil is almost always closer to the center than China and that Brazil is the one moving around.

The question one wonders after seeing the graphs is: why Brazil was getting closer over time? The main hypothesis discussed in the literature is the gravitational pull of the China economy, perhaps as

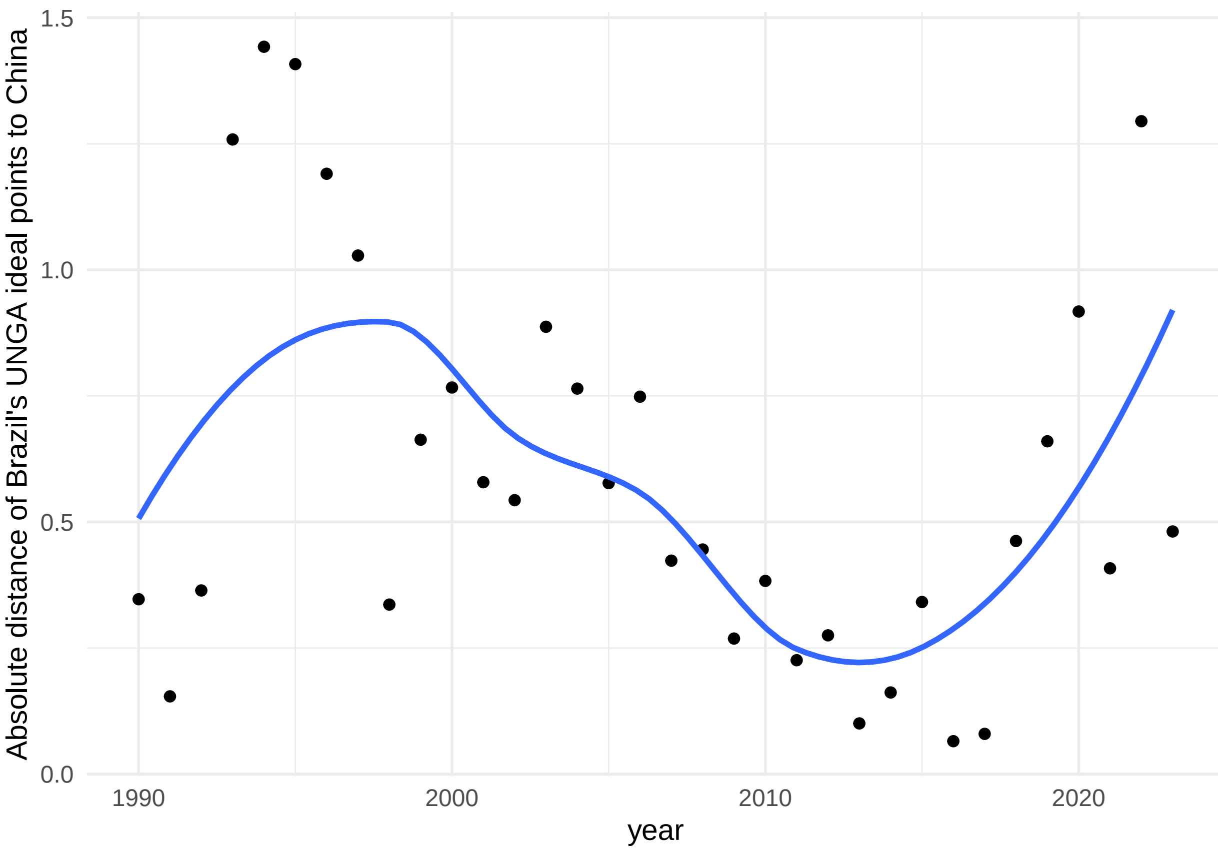

Figure 5: caption

shown in the percentage of exports of Brazil that went to China in the period considered. The figure below shows that around the year 2000, the percentage of exports destined to China increased at a roughly constant rate, in a very steep variation, which does not exactly fit the political alignment as measured by UNGA ideal points, although there is some positive correlation.

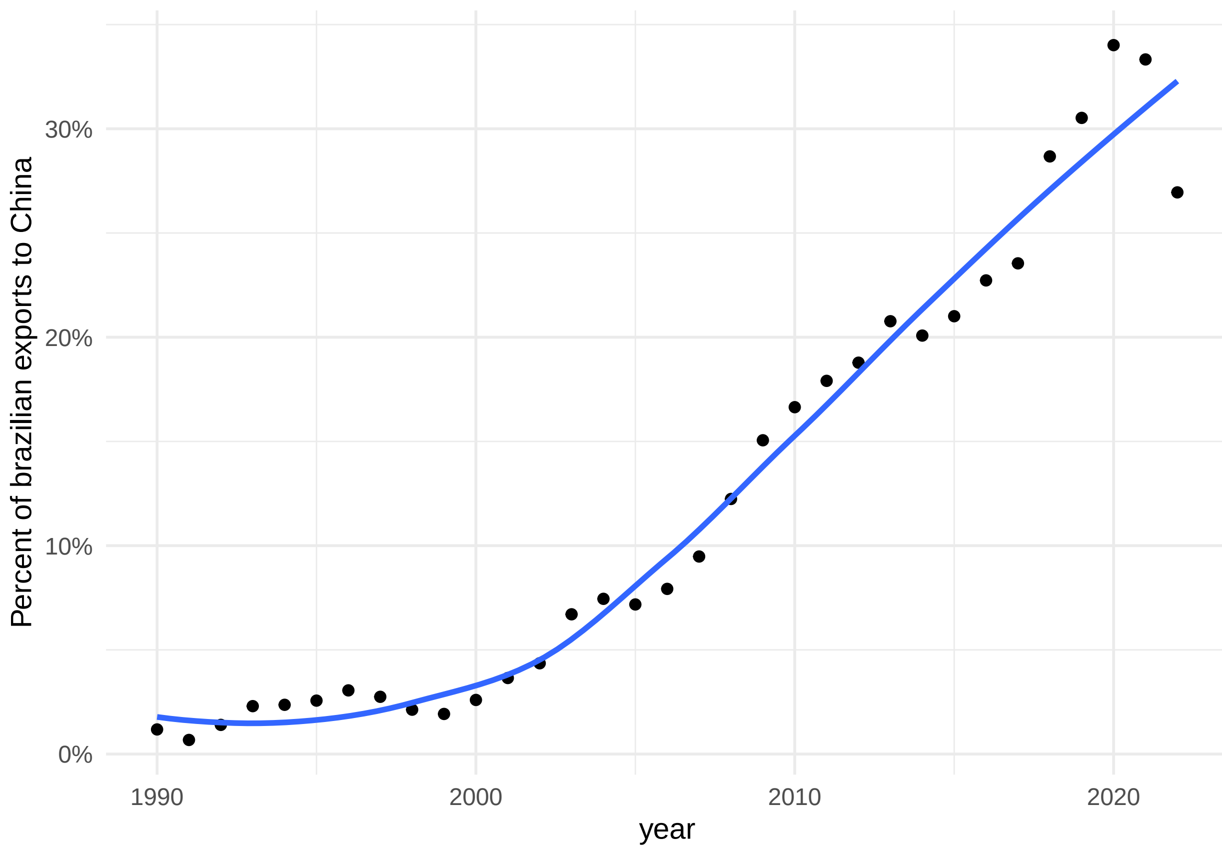

Figure 6: caption

6 Empirical Results

The graph below presents the evolution of political alignment between Brazil and China at UNGA and a donor pool of countries before and after China becoming the main trade partner of Brazil. The counterfactual estimate from the donor pool allows us to estimate the causal effect of the treatment on similarity of vote at UNGA. The estimate is -.27 points, which means that becoming the main trade partner implied a decrease of .27 points in the absolute distance between the ideal point of Brazil and China after 2006. The result is statistically significant at the 5% level, allowing us to reject the null hypothesis of no increase in the distance between the ideal points of both countries.

point estimate: -0.25 with 95% CI (-0.50, 0.01)

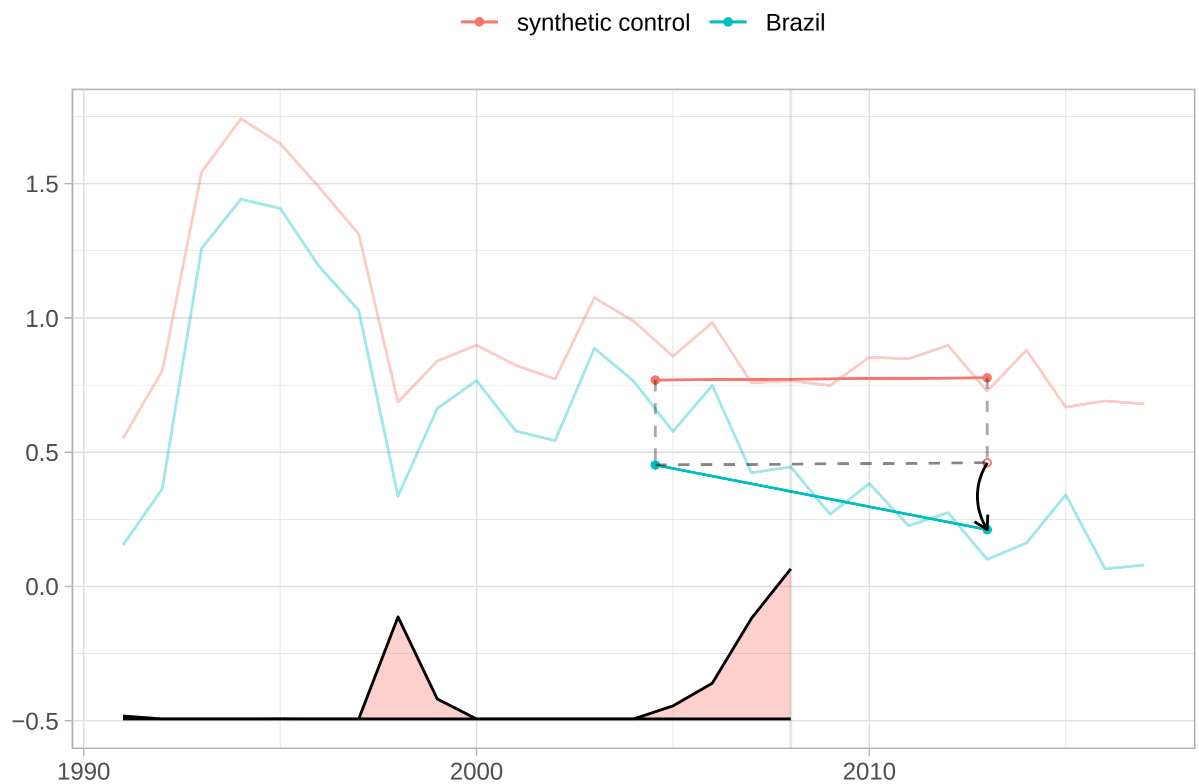

Figure 7: The blue line shows Brazil's change from the weighted pre-treatment average to the post-treatment average. The red one does the same for the synthetic control. The relative size of the time weights are in the bottom.

In figure 8 we plot the results fo Brazil along with top 10 controls accodring with their weights size. As we can see from the graph, no other country has a behavior similar to Brazil, which lends credibility to the estimation of the causal effect[2].

7 Robustness Checks

The target estimand of the SDiD estimator is the averate treatment effect on the treated for the period post treatment whose weights are different than zero. This means that, in theory, it may be capturing

[2]A figure with the weights for top controls is in the appendix

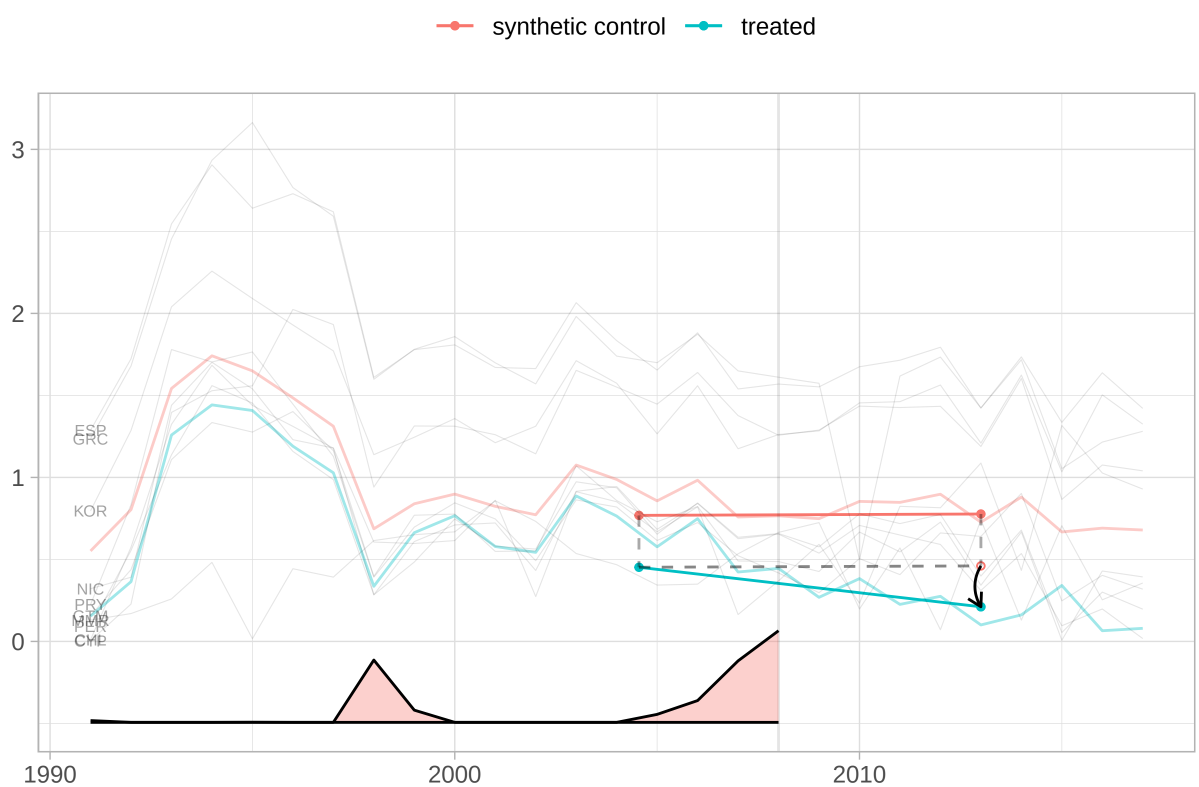

Figure 8: 10 controls with the largest weights

the effect of other treatments that happened in a later time. Another possibility is that the effect is from a change from before (say, the election of Lula in 2002, the first time a labour party assumed the presidency in Brazil) and we are capturing the the remanescent of such an effect. Thus, we ran two placebo checks, with the treatment occurring in 2003 and 2015, respectively. The table below reports the estimated effect along with the standard error. Our expectation is that the effect will be lower and non-significant, with the effect of the treatment in 2012 bigger (in absolute value) than in 2003, since the later is more distant in time from the true treatment.

Table 1: Synthetic DiD placebo tests (treatment years 2003 & 2012)

Model	Estimate	Std. Error	95 % CI lower	95 % CI upper
Placebo (2003)	-0.190	0.150	-0.485	0.104
Placebo (2012)	-0.148	0.146	-0.434	0.139

8 Conclusion

Building on insights from cognitive and attention-based theories, this paper is the first to use rank effect as an explanation for a major change in the foreign policy of a country. The symbolic salience of overtaking a historic trade partner prompted a substantive shift in diplomatic posture, above and beyond changes in trade volume.

We used a Synthetic Difference-in-Differences design to estimate that China's ascent to Brazil's top trade partner reduced their UNGA ideal-point absolute distance by 0.29 points (39%). Moreover, we provide evidence from a natural lanuage analysis from Brazilian main newspaper headlines to offer credible evidence that rank salience drives the alignment.

By spotlighting rank salience, this study opens new avenues for understanding how discrete milestones in economic ties can reshape international alignments—and invites extensions to other domains (FDI, diplomatic visits) and regions.

9 References

10 Appendix

Below we present the weights for the top 10 controls.

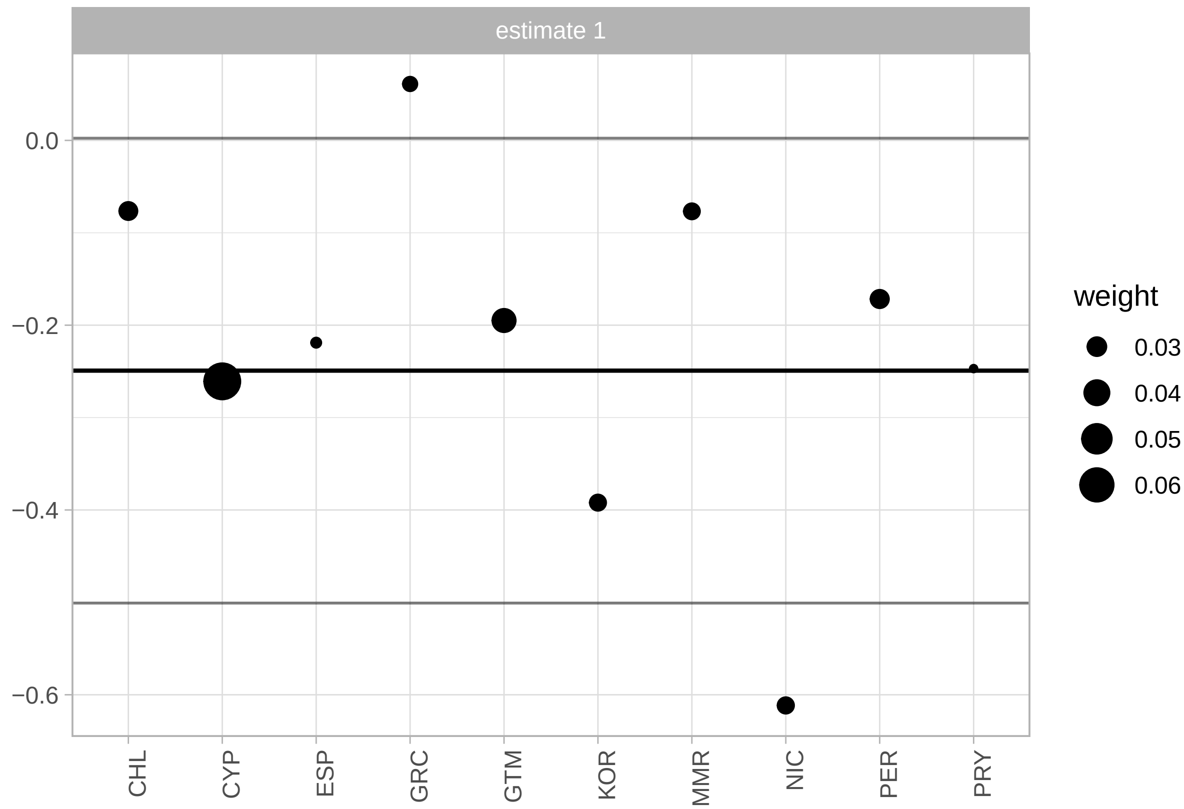

Figure 9: Weights of countries which contributed more to the construction of the synthetic control. The size of the dot represents the weight.

Aiyar, Shekhar, Davide Malacrino, and Andrea F. Presbitero. 2024. "Investing in Friends: The Role of Geopolitical Alignment in FDI Flows." *European Journal of Political Economy* 83 (June): 102508. https://doi.org/10.1016/j.ejpoleco.2024.102508.

Anagol, Santosh, and Thomas Fujiwara. 2016. "The Runner-up Effect." *Journal of Political Economy* 124 (4): 927–91.

Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. 2021. "Synthetic Difference-in-Differences." *American Economic Review* 111 (12): 4088–118. https://doi.org/10.1257/aer.20190159.

Bailey, Michael A, Anton Strezhnev, and Erik Voeten. 2017. "Estimating Dynamic State Preferences from United Nations Voting Data." *Journal of Conflict Resolution* 61 (2): 430–56.

Bastardoz, Nicolas, Michael J. Matthews, Gwendolin B. Sajons, Tyler Ransom, Thomas K. Kelemen, and Samuel H. Matthews. 2023. "Instrumental Variables Estimation: Assumptions, Pitfalls, and Guidelines." *The Leadership Quarterly* 34 (1): 101673. https://doi.org/10.1016/j.leaqua.2022.101673.

Blackwell, Matthew, and Adam N Glynn. 2018. "How to Make Causal Inferences with Time-Series Cross-Sectional Data Under Selection on Observables." *American Political Science Review* 112 (4): 1067–82.

Borchert, Ingo, Mario Larch, Serge Shikher, and Yoto V Yotov. 2021. "The International Trade and Production Database for Estimation (ITPD-E)." *International Economics* 166: 140–66.

Chen, Qin, and Yi Zhou. 2021. "Whose Trade Follows the Flag? Institutional Constraints and Economic Responses to Bilateral Relations." *Journal of Peace Research* 58 (6): 1207–23. https://doi.org/10.1177/0022343321992825.

Conlon, John J. 2023. "Attention, Neglect, and Persuasion."

Coutinho, Yuri Bravo, and Júlio César Cossio Rodriguez. 2024. "Chinese Double Effect on Brazilian Foreign Policy (2003-2018)." *Contexto Internacional* 46 (2): e20230014. https://doi.org/10.1590/s0102-8529.20244602e20230014.

Enke, Benjamin, Thomas Graeber, Ryan Oprea, and Jeffrey Yang. 2024. "Behavioral Attenuation." w32973. Cambridge, MA: National Bureau of Economic Research. https://doi.org/10.3386/w32973.

Enke, Benjamin, and Florian Zimmermann. 2017. "Correlation Neglect in Belief Formation." *The Review of Economic Studies*, December. https://doi.org/10.1093/restud/rdx081.

Feenstra, Robert C, Robert Inklaar, and Marcel P Timmer. 2015. "The Next Generation of the Penn World Table." *American Economic Review* 105 (10): 3150–82.

Flores-Macías, Gustavo A., and Sarah E. Kreps. 2013. "The Foreign Policy Consequences of Trade: China's Commercial Relations with Africa and Latin America, 1992–2006." *The Journal of Politics* 75 (2): 357–71. https://doi.org/10.1017/S0022381613000066.

Folke, Olle, Torsten Persson, and Johanna Rickne. 2016. "The Primary Effect: Preference Votes and Political Promotions." *American Political Science Review* 110 (3): 559–78.

Fuchs, Andreas, and Nils-Hendrik Klann. 2013. "Paying a Visit: The Dalai Lama Effect on International Trade." *Journal of International Economics* 91 (1): 164–77. https://doi.org/10.1016/j.jinteco.2013.04.007.

Granzier, Riako, Vincent Pons, and Clemence Tricaud. 2023. "Coordination and Bandwagon Effects: How Past Rankings Shape the Behavior of Voters and Candidates." *American Economic Journal: Applied Economics* 15 (4): 177–217. https://doi.org/10.1257/app.20210840.

Guilhon-Albuquerque, José-Augusto. 2014. "Brazil, China, US: A Triangular Relation?" *Revista Brasileira de Política Internacional* 57 (spe): 108–20. https://doi.org/10.1590/0034-7329201400207.

Haibin, Niu. 2010. "Emerging Global Partnership: Brazil and China." *Revista Brasileira de Política Internacional* 53: 183–92.

Hirshberg, David, and Sylvia Klosin. 2024. "Synthetic Differences-in-Differences with Covariates."

Imai, Kosuke, and In Song Kim. 2019. "When Should We Use Unit Fixed Effects Regression Models for Causal Inference with Longitudinal Data?" *American Journal of Political Science* 63 (2): 467–90. https://doi.org/10.1111/ajps.12417.

———. 2021. "On the Use of Two-Way Fixed Effects Regression Models for Causal Inference with Panel Data." *Political Analysis* 29 (3): 405–15.

Kastner, Scott L., and Margaret M. Pearson. 2021. "Exploring the Parameters of China's Economic Influence." *Studies in Comparative International Development* 56 (1): 18–44. https://doi.org/10.1007/s12116-021-09318-9.

Lal, Apoorva, Mackenzie Lockhart, Yiqing Xu, and Ziwen Zu. 2024. "How Much Should We Trust Instrumental Variable Estimates in Political Science? Practical Advice Based on 67 Replicated Studies." *Political Analysis* 32 (4): 521–40.

Mellon, Jonathan. 2021. "Rain, Rain, Go Away: 194 Potential Exclusion-Restriction Violations for Studies Using Weather as an Instrumental Variable." *American Journal of Political Science*.

Mintz, Alex. 2004. *The Poliheuristic Theory of Foreign Policy Decision Making*. Sage.

———. 2005. "Applied Decision Analysis: Utilizing Poliheuristic Theory to Explain and Predict Foreign Policy and National Security Decisions." *International Studies Perspectives* 6 (1): 94–98. https://doi.org/10.1111/j.1528-3577.2005.00195.x.

Morgan, Pippa. 2019. "Can China's Economic Statecraft Win Soft Power in Africa? Unpacking Trade, Investment and Aid." *Journal of Chinese Political Science* 24 (3): 387–409. https://doi.org/10.1007/s11366-018-09592-w.

Mourón, Fernando, and Francisco Urdinez. 2014. "A Comparative Analysis of Brazil's Foreign Policy Drivers Towards the USA: Comment on Amorim Neto (2011)." *Brazilian Political Science Review* 8 (2): 94–115. https://doi.org/10.1590/1981-38212014000100013.

Neto, Octavio Amorim. 2011. *De Dutra a Lula: A Condução e Os Determinantes Da Política Externa Brasileira*. Elsevier Brasil.

Reilly, James. 2013. "China's Economic Statecraft: Turning Wealth into Power."

Schenoni, Luis L., and Diego Leiva. 2021. "Dual Hegemony: Brazil Between the United States and China." In *Hegemonic Transition*, edited by Florian Böller and Welf Werner, 233–55. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-74505-9_12.

Struver, G. 2014. "'Bereft of Friends'? China's Rise and Search for Political Partners in South America." *The Chinese Journal of International Politics* 7 (1): 117–51. https://doi.org/10.1093/cjip/pot018.

Tanner, Murray Scot. 2007. *Chinese Economic Coercion Against Taiwan: A Tricky Weapon to Use*. Rand Corporation.

U.S. Department of Defense. 2018. "Assessment of U.S. Defense Implications of China's Expanding Global Access."

Vadell, Javier. 2011. "A China Na América Do Sul e as Implicações Geopolíticas Do Consenso Do Pacífico." *Revista de Sociologia e Política* 19: 57–79.

Yan, Jiaqiang, and Yonghong Zhou. 2021. "Economic Return to Political Support: Evidence from Voting on the Representation of China in the United Nations." *Journal of Asian Economics* 75 (August): 101325. https://doi.org/10.1016/j.asieco.2021.101325.

Characteristic	**N = 3,294**[1]
year	
Mean (SD)	2,004.00 (7.79)
Min, Max	1,991.00, 2,017.00
Population, millions	
Mean (SD)	47.15 (158.76)
Min, Max	0.10, 1,409.52
abs_distance_china	
Mean (SD)	0.82 (0.80)
Min, Max	0.00, 4.79
gdp_cur	
Mean (SD)	527,136.76 (1,679,733.14)
Min, Max	440.34, 17,797,724.00
ideal_point_all	
Mean (SD)	-0.18 (0.89)
Min, Max	-2.07, 3.22
abs_distance_usa	
Mean (SD)	2.97 (0.90)
Min, Max	0.00, 5.18
gpi	
Mean (SD)	0.01 (0.03)
Min, Max	0.00, 0.32
us_power_gap	
Mean (SD)	0.29 (0.04)
Min, Max	0.00, 0.32
us_ideal	
Mean (SD)	2.79 (0.21)
Min, Max	2.49, 3.22
trade_with_china	
Mean (SD)	5,067.57 (19,216.95)
Min, Max	0.00, 208,705.87
trade_with_us	
Mean (SD)	13,028.46 (44,742.64)
Min, Max	0.00, 535,592.25
total_trade	
Mean (SD)	84,190.26 (238,516.88)
Min, Max	0.05, 2,634,023.75
distance_us	
Mean (SD)	9,575.97 (3,681.06)
Min, Max	1,918.27, 16,413.34
distance_china	
Mean (SD)	9,562.53 (4,251.79)
Min, Max	1,214.74, 18,965.79
Official exchange rate (LCU per US$, period average)	
Mean (SD)	566.78 (2,247.11)
Min, Max	0.00, 33,226.30
pci_cur	
Mean (SD)	14,939.80 (17,746.92)
Min, Max	417.75, 151,188.10
perc_trade_with_china	